PRECALCULUS · UNIT 5: TRIGONOMETRY OF GENERAL TRIANGLES

Search and Rescue

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students coordinate a mountain rescue. Angles from two ranger towers locate a flare with the Law of Sines, team positions on bearings give distances with the Law of Cosines, and triangle area formulas divide the search region.

The Honors ambiguous-case radio fix shows why SSA needs extra information.

Standards Addressed

- G-SRT.C.8 Right-triangle trig in applied problems
- G-SRT.D.9 Area formula $\frac{1}{2}ab\sin C$
- G-SRT.D.10–11 Laws of Sines and Cosines
- Bearings and navigation

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a rescue news story. Part 1 elevations and the flare.
Day 2	Part 2 team positions and bearings.
Day 3	Part 3 search areas and the brief.

Materials

- Student handout and Field Data Cards
- Scientific calculators (degree mode)
- Protractors and rulers
- Optional: topographic map

How the Three Versions Differ

Level	What changes
CP2	Right-triangle trig, Law of Sines with the third angle, Law of Cosines for SAS, and area.
CP1	Bearings, helicopter time, team distances, Heron's formula, and dividing areas.

Level	What changes
Honors	Ambiguous case, coordinate navigation, landing site, verification with the Law of Cosines, and medians.

Watch For (Common Misconceptions)

- Calculator in radian mode.
- Pairing a side with the wrong angle in the Law of Sines.
- Forgetting the second triangle in SSA.

Extension Ideas

- Plot the rescue on a real map.
- Research how GPS uses triangulation.
- Create a rescue scenario for classmates.

Answer Key – CP2

Part 1: Tower Heights

#	Task	Answer
1	From 200 m away, the angle of elevation to the top of Tower A is 14°. How tall is the tower?	h = 200 tan 14° ≈ 49.9 m
2	A ranger in Tower A (50 m up) sees a tent at an angle of depression of 5°. How far away is the tent horizontally?	$\frac{50}{\tan 5°}$ ≈ 571.5 m
3	A rope from the top of a 30 m cliff to the ground makes a 62° angle with the ground. How long is the rope?	$\frac{30}{\sin 62°}$ ≈ 34.0 m
4	Explain the difference between angles of elevation and depression.	Elevation is measured up from horizontal; depression is down from horizontal.

Part 2: Find the Flare

#	Task	Answer
5	Find the angle at the flare.	180 − 52 − 61 = 67°
6	Use the Law of Sines to find the distance from Tower A to the flare.	AF = $\frac{8 \sin 61°}{\sin 67°}$ ≈ 7.6 km
7	Find the distance from Tower B to the flare.	BF = $\frac{8 \sin 52°}{\sin 67°}$ ≈ 6.8 km
8	Which tower should send its team? Why?	Tower B; it is closer.

Part 3: Team Distances

#	Task	Answer
9	Two teams leave camp. One walks 5 km and the other 7 km, with 60° between their paths. How far apart are they?	√(25 + 49 − 35) = √39 ≈ 6.2 km
10	Find the area of the triangle formed by the camp and the two teams. Use $\frac{1}{2}ab \sin C$.	$\frac{1}{2}$(5)(7) sin 60° ≈ 15.2 km²
11	Draw and label a map of the search with all distances you found.	Answers vary.

Reflection

#	Prompt	Look for
R1	When do you need the Law of Sines or Cosines instead of SOH-CAH-TOA?	When the triangle is not a right triangle.
R2	Why did you find the third angle first?	The Law of Sines needs an angle opposite each side used.

Answer Key – CP1

Part 1: Locate the Flare

#	Task	Answer
1	Find AF and BF.	Angle F = 67°; AF ≈ 7.60 km, BF ≈ 6.85 km
2	Tower B is due east of A (bearing 090°). The flare is north of the line AB. What is the bearing from A to the flare?	090° − 52° = 038°
3	The helicopter starts at Tower A. How long will it take to reach the flare at 180 km/h (in minutes)?	$\frac{7.60}{180}$ h ≈ 0.042 h ≈ 2.5 min
4	How far north of the line AB is the flare?	AF sin 52° ≈ 5.97 km

Part 2: Team Positions

#	Task	Answer
5	Find the angle between Team 1 (040°) and Team 2 (115°) at A.	75°
6	Find the distance between Team 1 and Team 2.	$\sqrt{(5.2^2 + 6.8^2 - 2(5.2)(6.8)\cos 75°)} \approx 7.41$ km
7	Find the distance from Team 1 to the flare. (Angle at A between Team 1 and the flare: 40° vs. 38°.)	$\sqrt{(5.2^2 + 7.60^2 - 2(5.2)(7.60)\cos 2°)} \approx 2.41$ km
8	Which team should go to the flare?	Team 1 (≈ 2.4 km away)

Part 3: Search Areas

#	Task	Answer
9	Find the area of triangle A–Team 1–Team 2.	$\frac{1}{2}(5.2)(6.8)\sin 75° \approx 17.1$ km²
10	Find the area of triangle A–B–F using Heron's formula (sides 8, 7.60, 6.85).	$s \approx 11.23$; $A \approx \sqrt{(11.23 \times 3.23 \times 3.63 \times 4.38)} \approx 24.0$ km²
11	Split the search among 3 teams so each searches at most 10 km² of triangle ABF. Is that possible? Explain.	24.0 ÷ 3 = 8.0 km² per team; yes

Reflection

#	Prompt	Look for
R1	How do bearings help you find angles inside triangles?	The difference of two bearings from the same point is the angle between the paths.
R2	Why did you use the Law of Cosines for team distances?	You knew two sides and the included angle (SAS).

Answer Key – Honors

Part 1: Ambiguous Signals

#	Task	Answer
1	A beacon is 7 km from Tower A. Tower B is 9 km from A, and angle B is 40°. How many triangles are possible? Find the possible angles at the beacon.	$\sin F = \frac{9 \sin 40°}{7} \approx 0.826 \rightarrow F \approx 55.7°$ or 124.3°; two triangles
2	Find the two possible distances from B to the beacon.	A = 180 − 40 − F: 84.3° or 15.7° → BF = $\frac{7 \sin A}{\sin 40°} \approx 10.84$ km or 2.95 km
3	A second ranger says the beacon is more than 5 km from B. Which triangle is correct?	The one with BF ≈ 10.84 km
4	Explain why the SSA case can give two triangles but SAS cannot.	In SSA the side opposite the given angle can swing to two positions; SAS fixes the included angle.

Part 2: Navigation

#	Task	Answer
5	Find the bearing and distance from Team 2 to the flare.	Using coordinates from A: F = 7.60(sin 38°, cos 38°) ≈ (4.68, 5.99); T2 = 6.8(sin 115°, cos 115°) ≈ (6.16, −2.87). Vector ≈ (−1.48, 8.86) → about 8.99 km on a bearing of about 351°.
6	Team 1 walks to the flare at 4 km/h and Team 2 at 5 km/h. Which arrives first?	Team 1: 2.41 ÷ 4 ≈ 0.60 h; Team 2: 8.99 ÷ 5 ≈ 1.80 h. Team 1 arrives first.
7	The helicopter must land 1 km from the flare in a clearing at bearing 200° from the flare. Find the clearing's distance from Tower A.	Clearing ≈ F + (sin 200°, cos 200°) ≈ (4.34, 5.05) → about 6.66 km from A
8	Verify one answer using the Law of Cosines instead of coordinates.	AF = 7.60, FC = 1, angle at F between FA (bearing 218°) and FC (200°) = 18° → $AC^2 = 57.76 + 1 - 2(7.60)\cos 18° \approx 44.3$ → AC ≈ 6.66 km ✓

Part 3: Search Plan

#	Task	Answer
9	Find the area of triangle ABF two ways ($\frac{1}{2}ab \sin C$ and Heron).	$\frac{1}{2}(8)(7.60) \sin 52° \approx 24.0$ km² both ways
10	A median from F splits the triangle into two equal areas. Find the length of the median to AB.	Midpoint M of AB: $FM^2 = 7.60^2 + 4^2 - 2(7.60)(4)\cos 52° \approx 36.3$ → FM ≈ 6.0 km
11	Write the operations brief: assignments, routes with bearings, times, and areas.	Answers vary.

Reflection

#	Prompt	Look for
R1	How did you decide which of the two ambiguous triangles was real?	Extra information (a second distance) eliminated one solution.
R2	Compare coordinates and the Law of Cosines for finding distances.	Both work; coordinates are systematic with bearings, while the Law of Cosines is faster when an included angle is known.